

PREDICTING HEART DISEASE RISK WITH MACHINE LEARNING

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PROBLEM DEFINITION & GOALS

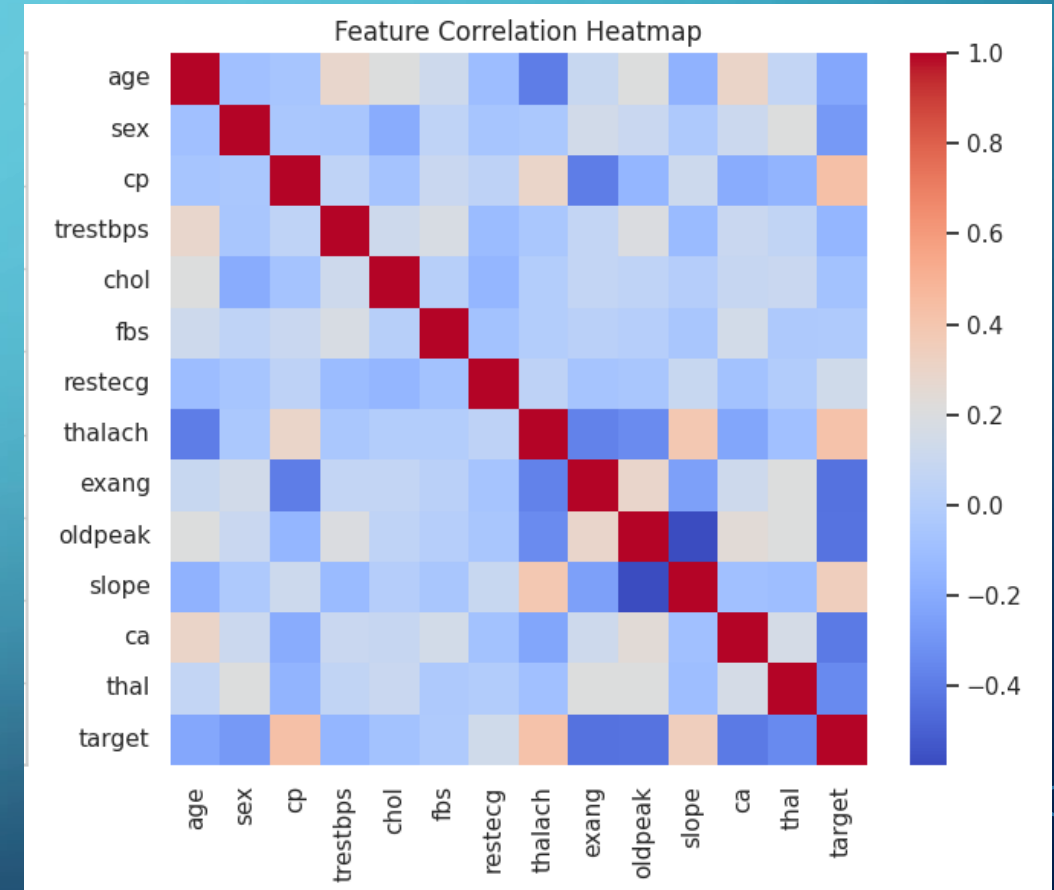
- **The Context:** Cardiovascular diseases are a leading cause of global mortality.
- **The Problem:** Early and accurate diagnosis is critical but complex.
- **Our Goal:** Build a reliable Machine Learning pipeline to predict heart disease risk based on patient clinical data.
- **Why it matters:** Assisting doctors in making faster, data-driven decisions to save lives.

DATASET DESCRIPTION & CLEANING

- **Source:** Public dataset from Kaggle (*johnsmith88*).
- **Initial Size:** 1025 rows, 14 columns (features like age, sex, cholesterol).
- **Target Variable:** Binary (0 = Healthy, 1 = Heart Disease).
- **Data Cleaning (Crucial Step):**
 - Found and removed **723 duplicate rows** to prevent model overfitting.
 - **Final Size:** 302 unique patients.
 - **Missing values:** 0.

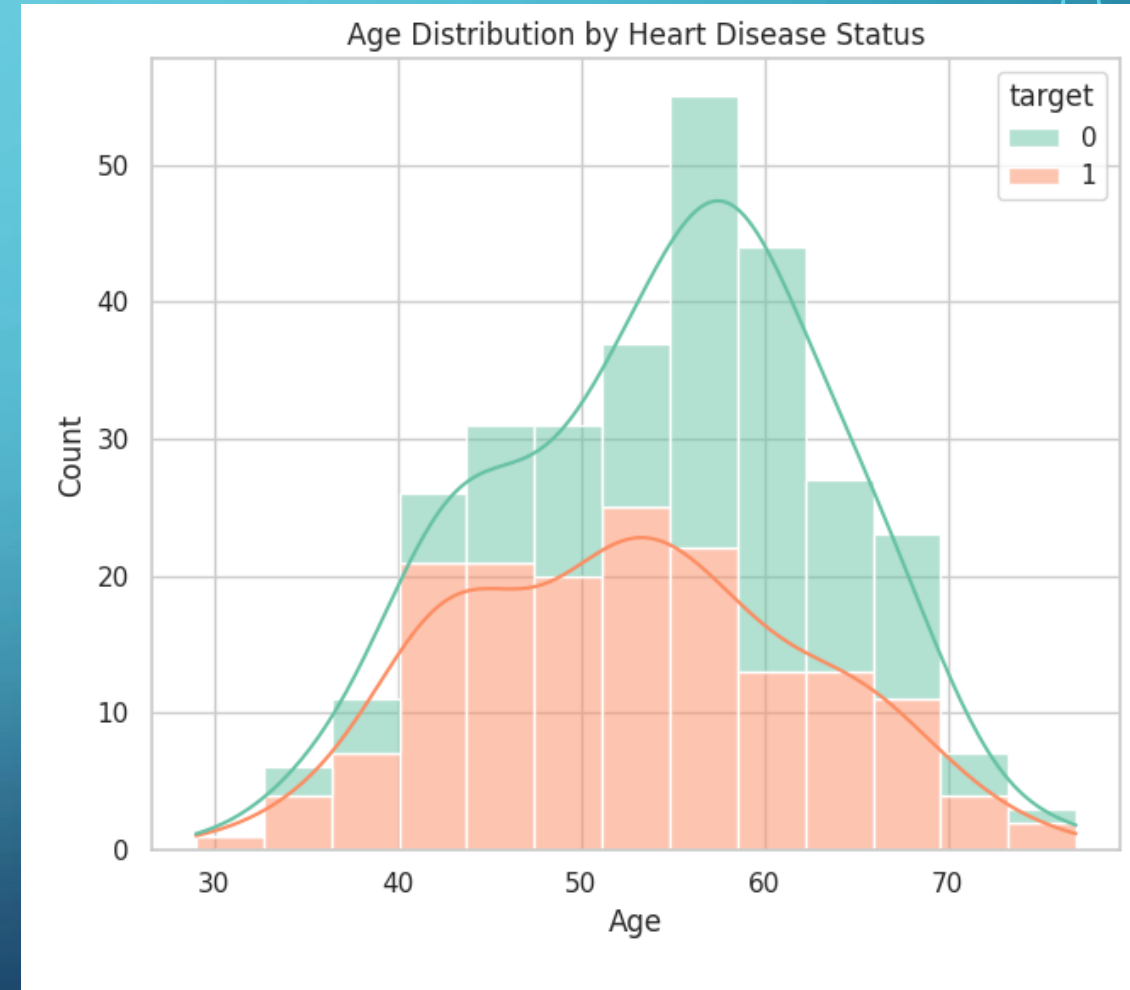
EXPLORATORY DATA ANALYSIS

- **Positive Correlations:** Chest pain type (cp) and maximum heart rate (thalach) are strong indicators of disease risk.
- **Negative Correlations:** Exercise-induced angina (exang) is inversely related to the target.



EXPLORATORY DATA ANALYSIS

- Disease risk is highly present across multiple age groups, not just the elderly.



FEATURE ENGINEERING & PREPROCESSING

- **Train-Test Split:**
 - 80% Training Data / 20% Testing Data.
 - `random_state=42` used for reproducibility.
- **Feature Scaling:**
 - Applied `StandardScaler` to normalize numerical features.
 - *Why?* Distance-based algorithms like KNN and Logistic Regression require scaled data to function correctly.

MACHINE LEARNING MODELS

- We trained and compared three classification algorithms:
- **Logistic Regression:** A strong baseline for binary medical classification.
- **K-Nearest Neighbors (KNN):** Looks at similar patient profiles ($n_neighbors=5$).
- **Random Forest:** An ensemble method handling complex, non-linear relationships.

MODEL COMPARISON & ACCURACY

- **Initial Accuracy Scores (on Test Data):**
 - KNN: 73.77%
 - Logistic Regression: 77.05%
 - Random Forest: **83.61%**
- *Conclusion:* Random Forest is our best-performing model.

DETAILED EVALUATION – WHY ACCURACY ISN'T ENOUGH

- **The Medical Reality:** A False Negative (telling a sick patient they are healthy) is extremely dangerous.
- **Our Results (Random Forest):**
 - **True Positives:** 26
 - **False Negatives:** Only 3
 - **Recall (Sensitivity):** 90% for detecting the disease

Random Forest Confusion Matrix

True label	Healthy (0)	Disease (1)
Healthy (0)	True Negative (TN) 25	False Positive (FP) 7
Disease (1)	False Negative (FN) 3	True Positive (TP) 26
	Healthy (0)	Disease (1)
	Predicted label	

RESPONSIBLE AI PRACTICES

- **Data Reliability:** Sourced from Kaggle, but originally from clinical databases (Cleveland).
- **Potential Biases:** Must ensure demographic groups (age, gender) are equally represented so the model doesn't favor one profile.
- **Real-World Risks:** Algorithms should *assist* doctors, not replace them. Blindly trusting AI without human verification can lead to fatal misdiagnoses.

CONCLUSION & FUTURE WORK

- **Summary:** Successfully built a pipeline from messy data to a highly accurate Random Forest predictor (83.6%).
- **Key Takeaways:** Data cleaning (duplicates) completely changed the reliability of our results.
- **Future Improvements:** * Hyperparameter tuning (GridSearch).
- Testing with a larger, more diverse real-world medical dataset.



THANK YOU FOR YOUR ATTENTION !